

# OE-EC704B — Optimization Technique

Semester VII • 3L:0T:0P • 3 Credits

## Module 2 Exam Notes: Single-Variable Optimization Algorithms

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## 1. Optimality Criteria for Single-Variable Functions

Optimality criteria determine whether a candidate point  $x^*$  is a local minimum, local maximum, or an inflection point.

### 1.1 Necessary Condition (First-Order)

#### Theorem: Necessary Condition

If a function  $f(x)$  is continuous and differentiable at  $x^*$ , and  $x^*$  is a local extremum (minimum or maximum), then the first derivative at  $x^*$  must vanish:

$$f'(x^*) = \left. \frac{df}{dx} \right|_{x=x^*} = 0 \quad (1)$$

Such a point  $x^*$  is called a **Stationary Point**.

### 1.2 Sufficient Condition (Second-Order)

#### Theorem: Sufficient Condition

Let  $f(x)$  be twice differentiable at the stationary point  $x^*$ .

- If  $f''(x^*) > 0$ , then  $x^*$  is a **local minimum**.
- If  $f''(x^*) < 0$ , then  $x^*$  is a **local maximum**.
- If  $f''(x^*) = 0$ , the test is inconclusive. We must evaluate higher-order derivatives:

$$f'(x^*) = f''(x^*) = \dots = f^{(n-1)}(x^*) = 0, \quad f^{(n)}(x^*) \neq 0 \quad (2)$$

- If  $n$  is **odd**,  $x^*$  is an **inflection point** (neither minimum nor maximum).
- If  $n$  is **even** and  $f^{(n)}(x^*) > 0$ ,  $x^*$  is a **local minimum**.
- If  $n$  is **even** and  $f^{(n)}(x^*) < 0$ ,  $x^*$  is a **local maximum**.

## 2. Bracketing Methods

Bracketing methods find an initial interval of uncertainty containing the optimum before applying fast search routines.

### 2.1 Exhaustive Search Method

Exhaustive search evaluates a function at equally spaced points in an interval  $[a, b]$  to find the sub-interval containing the minimum.

#### Exhaustive Search Algorithm

Let the initial interval be  $[a, b]$ , and let the number of evaluations be  $n$ .

1. Calculate the step size:

$$\Delta x = \frac{b - a}{n} \quad (3)$$

2. Define evaluation coordinates:  $x_i = a + i \cdot \Delta x$ , for  $i = 0, 1, \dots, n$ .
3. Evaluate  $f(x_i)$  sequentially for  $i = 1, 2, \dots, n - 1$ .
4. Identify three consecutive points  $x_{k-1}$ ,  $x_k$ , and  $x_{k+1}$  such that:

$$f(x_{k-1}) > f(x_k) \quad \text{and} \quad f(x_k) < f(x_{k+1}) \quad (4)$$

5. The minimum lies bracketed in the interval  $[x_{k-1}, x_{k+1}]$ . The new interval of uncertainty is:

$$L_1 = 2 \cdot \Delta x = \frac{2(b - a)}{n} \quad (5)$$

### 3. Region-Elimination Methods

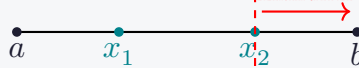
Region-elimination methods systematically reduce the interval of uncertainty by comparing objective values at test points.

#### 3.1 Working Principle of Region Elimination

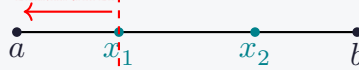
If we assume  $f(x)$  is **unimodal** (has only one minimum in the interval  $[a, b]$ ), we can eliminate a portion of the interval by evaluating two test points  $x_1$  and  $x_2$  (where  $x_1 < x_2$ ):

##### Unimodal Region Elimination Logic

If  $f(x_1) < f(x_2)$ , minimum cannot lie in  $(x_2, b]$



If  $f(x_1) > f(x_2)$ , minimum cannot lie in  $[a, x_1)$



#### 3.2 Interval Halving Method

This method eliminates exactly half of the interval at each stage using three evaluations ( $x_0$ ,  $x_1$ , and  $x_2$ ).

##### Interval Halving Algorithm

Let the current interval be  $[a, b]$ .

1. Calculate the midpoint:  $x_0 = \frac{a+b}{2}$ .
2. Calculate quarter points:

$$x_1 = a + \frac{b - a}{4}, \quad x_2 = b - \frac{b - a}{4} \quad (6)$$

3. Evaluate  $f(x_1)$ ,  $f(x_0)$ , and  $f(x_2)$ .
4. Compare function values to eliminate region:
  - If  $f(x_1) < f(x_0)$ , the new interval is  $[a, x_0]$ .
  - If  $f(x_2) < f(x_0)$ , the new interval is  $[x_0, b]$ .
  - If  $f(x_1) \geq f(x_0)$  and  $f(x_2) \geq f(x_0)$ , the new interval is  $[x_1, x_2]$ .
5. In all cases, the new interval width is exactly:

$$L_{new} = \frac{b - a}{2} \quad (7)$$

### 3.3 Fibonacci Search Method

Fibonacci search is the mathematically optimal region-elimination method. It uses the Fibonacci sequence  $F_0 = 1, F_1 = 1, F_n = F_{n-1} + F_{n-2}$  to calculate placement ratios.

#### Fibonacci Search Equations

For a pre-selected total number of function evaluations  $n$ :

1. The initial two test points  $x_1$  and  $x_2$  in  $[a, b]$  are placed at:

$$x_1 = a + \frac{F_{n-2}}{F_n}(b - a) \quad (8)$$

$$x_2 = b - \frac{F_{n-2}}{F_n}(b - a) = a + \frac{F_{n-1}}{F_n}(b - a) \quad (9)$$

2. After region elimination, the remaining sub-interval is evaluated. At the  $k$ -th step, the new test point is placed at:

$$x_k = \text{Remaining Limit} \pm \frac{F_{n-k-1}}{F_{n-k+1}}(\text{Current Interval Width}) \quad (10)$$

3. The final reduction ratio of the interval of uncertainty after  $n$  iterations is:

$$\frac{L_n}{L_1} = \frac{1}{F_n} \quad (11)$$

## 4. Point Estimation Methods

Unlike region elimination, point estimation methods fit a smooth polynomial through known points to estimate the local minimum directly.

### 4.1 Successive Quadratic Estimation (SQE)

SQE fits a quadratic function  $q(x) = Ax^2 + Bx + C$  through three test points  $x_1 < x_2 < x_3$  with known function values  $f_1, f_2, f_3$ .

**SQE Minimum Point Formula**

The minimum  $x^*$  of the approximating quadratic polynomial is given by:

$$x^* = \frac{1}{2} \frac{(x_2^2 - x_3^2)f_1 + (x_3^2 - x_1^2)f_2 + (x_1^2 - x_2^2)f_3}{(x_2 - x_3)f_1 + (x_3 - x_1)f_2 + (x_1 - x_2)f_3} \quad (12)$$

Alternatively, we can express the calculation using coordinate differences:

$$x^* = x_2 + \frac{1}{2} \frac{(x_2 - x_1)^2(f_2 - f_3) - (x_2 - x_3)^2(f_2 - f_1)}{(x_2 - x_1)(f_2 - f_3) - (x_2 - x_3)(f_2 - f_1)} \quad (13)$$

If  $x^*$  is close to  $x_2$  within a tolerance limit,  $x^*$  is taken as the optimum. Otherwise, the worst point among  $\{x_1, x_2, x_3\}$  is replaced by  $x^*$ , and the calculation is repeated.

## Quick Revision — Module 2

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### Key Formulas & Equations

|                                  |                                                                                                                                         |
|----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| <b>Necessary Condition</b>       | $f'(x^*) = 0$ (stationary point requirement).                                                                                           |
| <b>Sufficient Condition</b>      | $f''(x^*) > 0 \Rightarrow$ local min; $f''(x^*) < 0 \Rightarrow$ local max.                                                             |
| <b>Interval Halving Width</b>    | $L_{new} = \frac{1}{2}L_{old}$ (achieved with 3 evaluations).                                                                           |
| <b>Fibonacci Point 1</b>         | $x_1 = a + \frac{F_{n-2}}{F_n}(b - a)$ .                                                                                                |
| <b>Fibonacci Reduction Ratio</b> | $\frac{L_n}{L_1} = \frac{1}{F_n}$ (where $F_n$ is the $n$ -th Fibonacci number).                                                        |
| <b>SQE Minimum Point</b>         | $x^* = x_2 + \frac{1}{2} \frac{(x_2 - x_1)^2(f_2 - f_3) - (x_2 - x_3)^2(f_2 - f_1)}{(x_2 - x_1)(f_2 - f_3) - (x_2 - x_3)(f_2 - f_1)}$ . |

### One-Line Definitions

- **Stationary Point:** A point at which the first derivative of a differentiable function is zero.
- **Inflection Point:** A point on a curve at which the concavity changes, where  $f''(x) = 0$  and  $f'''(x) \neq 0$ .
- **Unimodal Function:** A function that has exactly one local extremum (minimum or maximum) in a given interval.
- **Bracketing Method:** A search algorithm designed to find an initial interval containing the optimum.
- **Region Elimination:** Systematically discarding a sub-interval of search space using unimodality comparisons.
- **Fibonacci Search:** An optimal region-elimination search algorithm utilizing Fibonacci ratios.
- **Successive Quadratic Estimation:** An interpolation algorithm that fits quadratic curves through test coordinates to locate the minimum.

### Frequently Asked / Viva Questions

**Q1. State the necessary and sufficient conditions for a local minimum of a single-variable function.**

Necessary condition:  $f'(x^*) = 0$ . Sufficient condition:  $f''(x^*) > 0$ .

**Q2. What happens if  $f'(x^*) = f''(x^*) = 0$  at a candidate point?**

The test is inconclusive. We must evaluate higher-order derivatives. If the first non-zero derivative is of odd order, it is an inflection point. If it is of even order and positive, it is a local minimum.

## Frequently Asked / Viva Questions (contd.)

**Q3. Define a unimodal function and state why it is important for region-elimination methods.**

A function is unimodal if it has only one optimum in a given range. Unimodality guarantees that comparing function values at two points ( $f(x_1)$  and  $f(x_2)$ ) yields a valid sub-region to eliminate.

**Q4. How does the Interval Halving method work?**

It divides the interval into four equal parts using three points ( $x_1 < x_0 < x_2$ ). Comparing their function values allows the elimination of half the interval ( $L_{new} = 0.5L_{old}$ ).

**Q5. Why is the Fibonacci search method considered optimal among region-elimination techniques?**

For a pre-specified number of evaluations  $n$ , it achieves the maximum possible reduction in the interval of uncertainty ( $1/F_n$ ) without requiring any derivative information.

**Q6. What is the relation between Fibonacci search and Golden Section search?**

Golden Section search is the limiting case of Fibonacci search as the number of evaluations  $n \rightarrow \infty$ . It uses the constant Golden Ratio  $\gamma \approx 0.618$  instead of discrete Fibonacci fractions.

**Q7. Write down the quadratic estimation formula for the minimum point  $x^*$ .**

$$x^* = \frac{1}{2} \frac{(x_2^2 - x_3^2)f_1 + (x_3^2 - x_1^2)f_2 + (x_1^2 - x_2^2)f_3}{(x_2 - x_3)f_1 + (x_3 - x_1)f_2 + (x_1 - x_2)f_3} \text{ for three ordered points } x_1 < x_2 < x_3.$$

**Q8. Differentiate between bracketing methods and point estimation methods.**

Bracketing methods find an interval containing the optimum (e.g., Exhaustive Search). Point estimation methods fit functional approximations (e.g., quadratic curves) to predict the exact coordinate of the optimum.

**Q9. Under what conditions does the successive quadratic estimation (SQE) method converge?**

SQE converges rapidly if the objective function is smooth, continuous, and has a unimodal shape resembling a parabola in the neighborhood of the optimum.

**Q10. What is the significance of the Fibonacci numbers sequence in search algorithms?**

The Fibonacci numbers satisfy the property  $F_{n-2}/F_n \rightarrow 0.382$  and  $F_{n-1}/F_n \rightarrow 0.618$ , which balances the spacing of evaluation points so that previous calculations are reused in subsequent steps.